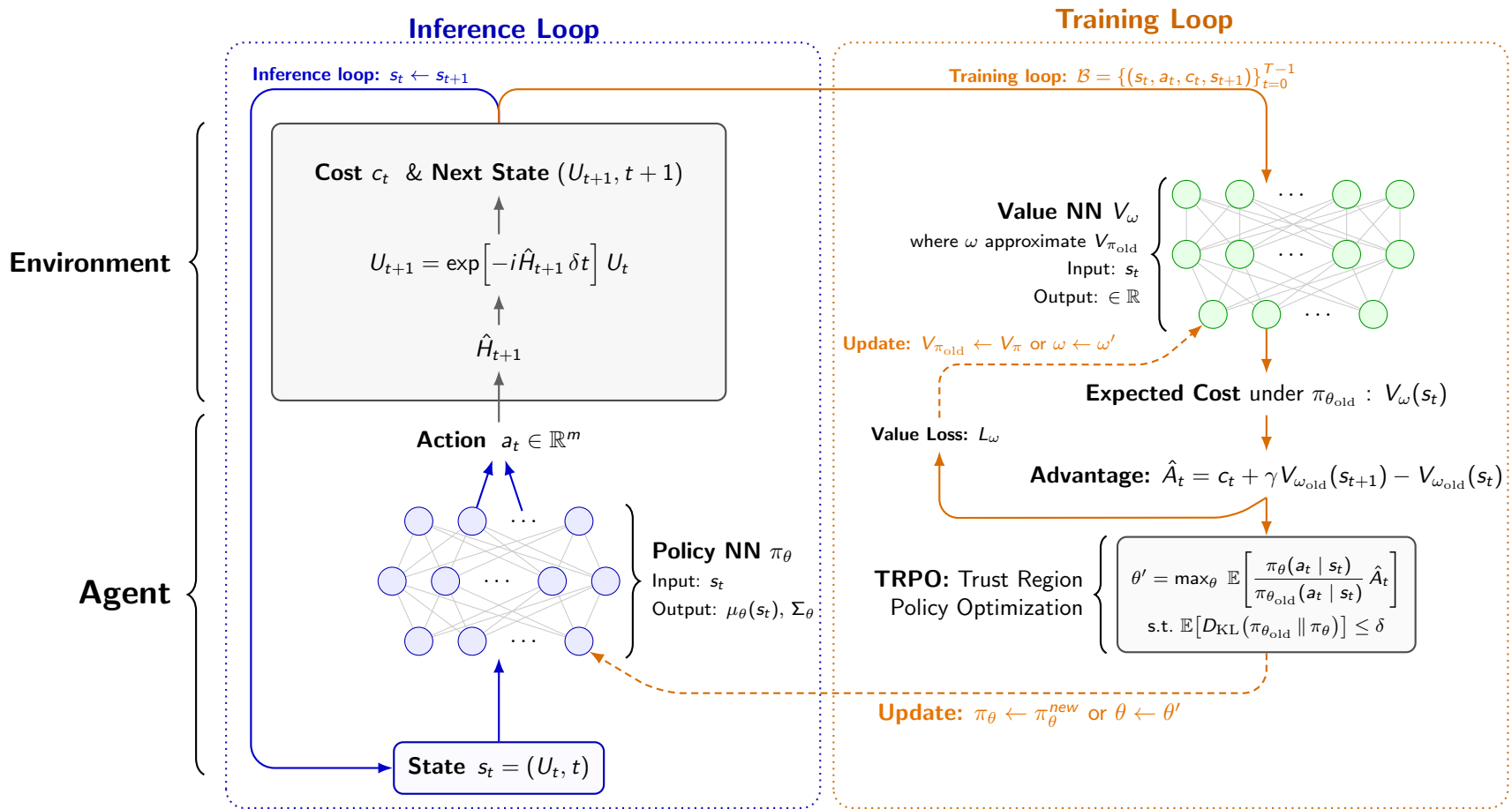
Inference Loop
Training Loop
Inference loop: $s_t \leftarrow s_{t+1}$
Training loop: $\mathcal{B} = \{(s_t, a_t, c_t, s_{t+1})\}_{t=0}^{T-1}$
Environment
Agent
Cost c_t & Next State $(U_{t+1}, t+1)$
$U_{t+1} = \exp\left[-i\hat{H}_{t+1}\,\delta t\right] U_t$
$\hat{H}_{t+1}$
Action $a_t \in \mathbb{R}^m$
Policy NN π_θ
Input: s_t
Output: $\mu_\theta(s_t)$, Σ_θ
State $s_t = (U_t, t)$
Value NN V_ω
where ω approximate $V_{\pi_{\text{old}}}$
Input: s_t
Output: $\in \mathbb{R}$
Update: $V_{\pi_{\text{old}}} \leftarrow V_\pi$ or $\omega \leftarrow \omega'$
Expected Cost under $\pi_{\theta_{\text{old}}}$: $V_\omega(s_t)$
Value Loss: L_ω
Advantage: $\hat{A}_t = c_t + \gamma V_{\omega_{\text{old}}}(s_{t+1}) - V_{\omega_{\text{old}}}(s_t)$
TRPO: Trust Region Policy Optimization
$\theta' = \max_\theta \mathbb{E}\left[\frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)} \hat{A}_t\right]$
s.t. $\mathbb{E}\left[D_{\text{KL}}(\pi_{\theta_{\text{old}}} \parallel \pi_\theta)\right] \leq \delta$
Update: $\pi_\theta \leftarrow \pi_\theta^{new}$ or $\theta \leftarrow \theta'$